D Kishor Kumar Reddy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.58
SNG VIDYA BHAWAN SECTOR 4 BHILAI	Class XII (CGBSE)	2016	54.80%
BSP SENIOR SECONDARY SCHOOL SECTOR10 BHILAI	Class X (CBSE)	2014	6.20%

SKILLS

Programming language : Python, MATLAB, SQL

CI/CD Tool : Jenkins, NPM, SonarQube, Nexus

Version control & SCM : Git- VCS ,GitHub -SCM

Configuration management : Docker, Kubernetes

Cloud platform : AWS EC2, S3, IAM, VPC, Lambda

Monitoring & Logging : Prometheus, Grafana (basic)

Infrastructure as code : Terraform

AI-ML Framework and Libraries : Gen AI, Agentic AI, Scikit-learn, TensorFlow, Keras, PyTorch

CAD-Modelling and Simulation : SOLIDWORKS, Fusion 360, Abaqus, Ansys

Data analysis and visualization : MS Excel , Pandas, NumPy, Matplotlib

Core CS- Data Structure and Algorithm, Object -Oriented Programming, System Design Fundamentals

PROJECTS

Mtech Project: **Residual Stress Evolution and Machine Learning-Based Prediction in Hybrid Deformation-Deposition Manufacturing of Non-Planar Components.**

Guide : prof. N Venkata Reddy

Effect of Non-Planar Process Parameters on Weld Bead Profile in WAAM

- Designed and conducted a **Box-Behnken Design of Experiments (DOE)** to study the effects of current, travel speed, relative angle, and deposition slope on weld bead geometry.
- Prepared samples through **polishing and etching**, captured bead profiles using stereo microscopy, and performed **regression analysis in Minitab**, with force prediction models validated through experimental results and analytical calculations.

Learning Management System | React.js, Node.js, PostgreSQL, Docker, Jenkins (GitHub)

- Built a three-tier Learning Management System using **React.js, Node.js, and PostgreSQL**, covering the frontend, backend services, and data management.
- Containerized the application with **Docker Compose** and implemented **Jenkins-based CI/CD** to automate the build and deployment process.

CFD Analysis of Fluid Flow and Heat Transfer in a Mixing Elbow

- Performed CFD simulations to study fluid flow and heat transfer in a mixing elbow, analyzing the effects of secondary-pipe injection position and angle on thermal mixing.
- Evaluated six configurations using temperature contours and velocity vectors, with mixing efficiency ranging from

66% to 81% across the simulated cases.

Thermomechanical Modelling of Residual Stress and Distortion in LPBF

- Developed an Abaqus-based thermomechanical model of an Inconel 718 LPBF cantilever and compared cooling from 800°C to 20°C with a preheated plate condition of 800°C to 300°C.
- Evaluated Von Mises stress and vertical displacement to study the effect of substrate preheating, observing reduced residual stress and warping under preheated conditions.

Courses Recommendation Engine **(Github)**

NextStep AI is a personalized course-path recommender that helps learners decide what to study next. A learner describes a goal, chooses a target knowledge level, and can optionally list completed modules. The app uses that information to generate distinct, ordered learning paths with course titles, descriptions, links, and a short explanation of how each path supports the goal.

POSITIONS OF RESPONSIBILITY

Teaching assistant(TA) under Prof. Venkata Reddy in the Mechanical and Aerospace Engineering department at IIT Hyderabad

Co-organiser, Yuva Fest, Uttaranchal University, Dehradun

Alumni Association Coordinator , Uttaranchal University, Dehradun

RELEVANT COURSES

Design & Simulation: Finite Element Method (FEM), Computational Fluid Dynamics (CFD), FEM Laboratory, Elasticity & Plasticity

Manufacturing: Additive Manufacturing, Design for Manufacturing & Assembly (DFMA), CAD/CAM, Industry 4.0, Scaling Laws & Multiscale Manufacturing, Advanced Material Joining Processes

AI/ML: Machine Learning and its Applications

TRAINING AND EXPERIENCE

SAIL (Steel Authority of India Limited)

Learned basic safety standards and production flow

Observed large-scale steel manufacturing processes

Simplex casting ltd.

Gained practical exposure to foundry and casting processes, including mould preparation, metal melting, pouring, solidification, and casting inspection.

NSPCL (NTPC-SAIL Power Corporation Limited)

Understanding of thermal power generation processes

Participated in routine maintenance observations

CERTIFICATIONS

Devops, Digital Edify, Hyderabad 2025

AWS Cloud, Digital Edify, Hyderabad 2025

Industrial safety, BSR Institute of Fire and Safety Engineering SBTET, Andhra Pradesh